

AKRON FULTON, OH, USA

WMO: 724303

Lat: 41.038N

Lon: 81.464W

Elev: 1044

StdP: 14.15

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 14813

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	3.1	8.5	-4.7	4.4	5.0	0.0	5.7	10.2	25.6	32.7	23.4	29.3	7.0	250	0.494

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.7	89.7	72.6	87.3	71.5	84.7	70.3	75.3	84.9	74.0	82.9	72.7	81.2	9.8	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
72.4	124.7	80.7	71.2	119.4	79.1	69.9	114.3	77.9	39.4	85.8	38.1	82.7	36.8	81.0	81.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.4	18.8	16.9	DB	-2.4	92.7	6.9	2.7	-7.4	94.7	-11.5	96.3	-15.4	97.8	-20.4	99.8
			WB	-3.1	77.8	6.6	1.4	-7.9	78.8	-11.7	79.6	-15.5	80.4	-20.2	81.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.4	27.2	29.4	38.2	50.6	61.2	69.3	73.4	72.1	65.5	53.6	42.2	32.8
	DBStd	18.66	11.90	11.37	11.21	9.60	8.71	6.52	5.37	5.24	7.39	8.94	9.16	10.18
	HDD50	2680	710	584	398	108	11	0	0	0	1	63	265	540
	HDD65	5798	1171	997	830	440	180	34	4	7	82	371	683	999
	CDD50	3196	4	7	33	127	359	580	724	684	466	173	33	6
	CDD65	838	0	0	1	8	63	164	263	226	97	16	0	0
	CDH74	6901	0	0	9	91	566	1324	2239	1775	790	105	2	0
	CDH80	2050	0	0	0	8	126	384	746	531	236	19	0	0
Wind	WSAvg	7.4	9.4	8.8	8.4	8.3	6.8	6.6	5.9	5.5	5.5	7.0	7.9	8.7
Precipitation	PrecAvg	39.2	2.8	2.1	2.8	3.4	4.0	3.5	4.4	3.9	3.6	3.1	2.8	2.8
	PrecMax	51.4	5.9	4.9	5.7	5.7	7.8	5.9	11.4	8.1	7.2	6.1	5.6	6.9
	PrecMin	25.6	1.4	1.0	0.5	1.6	1.6	0.5	1.2	1.1	1.2	1.1	1.4	0.9
	PrecStd	5.9	1.1	1.0	1.2	1.1	1.8	1.5	2.5	1.6	1.6	1.3	1.0	1.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	61.2	67.5	74.0	81.3	86.8	90.3	92.7	91.5	90.5	82.5	71.7	62.9
		MCWB	55.1	57.0	59.7	63.4	70.0	72.5	75.2	73.1	72.0	69.5	60.0	55.8
	2%	DB	56.1	58.2	66.9	76.6	83.4	87.6	89.7	88.4	86.5	76.6	65.5	57.0
		MCWB	52.4	51.9	55.9	60.6	67.9	71.3	73.3	72.0	70.9	64.8	56.9	51.6
	5%	DB	50.3	52.3	61.5	72.3	80.5	84.5	87.0	85.6	82.1	72.4	61.4	52.8
		MCWB	46.2	46.9	52.8	58.4	66.3	70.1	71.8	70.9	68.6	63.1	53.5	48.6
	10%	DB	44.7	46.0	56.8	67.8	76.8	81.5	84.2	82.7	78.6	68.4	57.0	47.7
		MCWB	40.6	40.6	49.5	55.6	64.2	68.2	70.4	69.6	66.9	60.5	50.8	44.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.8	58.3	61.0	66.5	73.0	75.4	78.0	76.8	75.4	71.3	62.9	57.6
		MCDB	59.9	65.0	71.1	77.2	84.2	86.4	88.6	87.0	86.4	80.6	68.1	60.8
	2%	WB	52.9	52.8	57.8	62.9	70.2	73.4	75.7	74.8	73.1	68.1	58.7	53.1
		MCDB	56.2	57.1	65.1	72.4	80.0	83.5	85.2	83.4	82.0	74.3	64.2	56.2
	5%	WB	46.7	46.8	54.4	60.3	68.2	71.8	74.3	73.3	71.0	64.5	54.8	48.6
		MCDB	49.7	52.5	60.2	69.6	77.3	81.4	83.2	81.0	78.5	70.3	59.9	51.9
	10%	WB	40.6	40.8	50.0	57.6	66.1	70.3	72.8	71.8	68.9	61.2	51.4	44.0
		MCDB	44.4	46.0	56.4	65.8	74.5	78.7	81.0	79.4	75.8	67.2	56.5	47.7
Mean Daily Temperature Range	5% DB	MDBR	13.7	15.5	17.5	20.4	20.2	18.8	18.7	18.6	19.2	17.6	15.5	12.4
		MCDBR	19.0	22.6	24.4	26.2	23.6	21.5	20.8	21.3	22.6	22.5	22.3	18.5
		MCWBR	16.5	17.7	15.8	15.3	11.8	10.1	8.4	8.6	9.9	12.8	15.4	15.6
	5% WB	MCDBR	17.9	21.0	21.9	22.4	20.3	18.8	17.5	17.0	17.9	18.6	19.5	17.7
		MCWBR	17.0	17.8	15.3	15.0	11.1	9.8	8.2	8.2	8.9	12.0	15.3	16.5
Clear-Sky Solar Irradiance	taub		0.301	0.329	0.356	0.389	0.427	0.449	0.450	0.439	0.384	0.362	0.319	0.303
	taud		2.425	2.330	2.338	2.281	2.200	2.178	2.193	2.215	2.343	2.398	2.493	2.462
	Ebn at Noon		271	278	283	280	270	264	262	262	270	263	263	261
	Edh at Noon		25	32	36	41	45	46	45	43	35	30	23	23
All-Sky Solar Radiation	RadAvg		452	730	1039	1420	1699	1866	1864	1677	1345	876	566	375
	RadStd		40	93	106	135	144	143	128	84	123	79	85	38

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	+2.34	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+484	+328
(47) Regional (50 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+202	+108

Nomenclature: See separate page